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Interspecific differences in the regeneration of artificial injuries on scleractinian corals

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Abstract

Routine injury of corals occurs frequently and is caused by natural processes and human activities. Morphological theory predicts that sessile, marine animals are strongly committed to their place of settlement, will invest more resources into maintenance and defence than species that are more fugitive in nature (Jackson, 1979). In this study, I conducted several field experiments to determine if the regenerative ability of injured colonies differed among species with different morphologies, and if regeneration was influenced by the position of the injury within the colony or the type of injury. Eleven species exhibited a wide range of responses to damage, from little to no change to complete recovery within 71 to 286 days. Regenerative ability could be ranked according to morphological attributes for these species (arborescent > bushy > tabular > massive > submassive), demonstrating that morphology does influence the recovery of injuries. Regeneration of different types of injury and injuries located at different positions within a colony also varied among species. Repair of scraping injuries was greater than tissue injuries, while regrowth of a new branch was slowest of all. With the exception of *Porites mayeri* which repaired more of its central injury than its edge injury, recovery rate of central and edge injuries was not significantly different for *Acropora robusta, A. hyacinthus, A. palifera, Pocillopora damcornis*, and *Porites lichen*. These results imply that the amount of damage caused by an injury differs, both for the species it is inflicted upon, and the type of injury generated. The ability to recover from injuries is important in corals since poor regenerative ability can lead to a reduction in colony fitness. © 1997 Elsevier Science B.V.

Keywords: Branch removal; Coral reefs; Damage; Impact; Injury; Life-history; Regeneration; Scleractinian corals; Scraping injury; Tissue removal

1. Introduction

Damage or injury to coral colonies originates from natural causes (e.g. predation,

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competition, storm and cyclone damage) or human activities such as anchoring, diving, mining and pollution (e.g. reviews by Connell, 1973; Brown and Howard, 1985; Craik et al., 1990; Grigg and Dollar, 1990). Corals have varying susceptibilities to damage by these processes due to morphological differences (Woodley et al., 1981; Hughes, 1989; Glynn, 1990; Chadwick-Furman, 1995). For example, branching species are more prone to storm and diver damage than massive species (Woodley et al., 1981; Chadwick-Furman, 1995), and massive and encrusting species are more vulnerable to overgrowth by algae (Hughes, 1989). Their ability to recover from damage is also variable since regeneration is energetically costly, and species differ in the allocation of resources between regeneration and other demographic processes (e.g. Bak et al., 1977; Bak and Steward-Van Es, 1980; Meesters and Bak, 1993; Meesters et al., 1993). Morphological theory of sessile, marine animals predicts that species with a greater commitment to their place of settlement will invest more resources towards defence and maintenance (Jackson, 1979). This hypothesis was based on eight shape parameters which characterised the different morphological strategies of sessile animals. These parameters included tissue area and volume (potential for feeding, respiration and reproduction, etc.), skeletal volume (measure of support material and strength), substratum utilisation, directional growth (vertical or horizontal), substratum interactions (pre-emption of space, peripheral encounters), and holes (complexity of form in determining the availability of habitats for other organisms). The impact that an injury has on a colony will be directly related to these parameters, particularly tissue and skeleton volume and tissue area. The first objective of this study is to determine if morphology does influence rates of regeneration in corals by monitoring the recovery of artificial lesions for a range of species with contrasting shapes.

The recovery of injuries in corals is a complex process that is intrinsically related to the characteristics of the injury (e.g. type and position within the colony: Bak et al., 1977; Bak and Steward-Van Es, 1980; Meesters et al., 1993; Meesters and Bak, 1995). Routine damage in corals results in the removal of tissue, or tissue and skeleton from a colony. The amount of loss which occurs for these two injury types can be further categorised as partial or total for tissue injuries and superficial or extensive for tissue and skeleton injuries. A wide range of causative agents generate these different types of injuries. Generally, extensive tissue loss results in the death of polyps within a given area and is due to predation (e.g. *Acanthaster plancii*), sedimentation, and solar irradiance (amongst other factors) while the partial removal of polyps (partial tissue loss) is caused by the action of polyp grazers such as some Chaetodontid fish (Brown, 1988, 1994; Glynn, 1990; Stafford-Smith, 1993). The superficial loss of tissue and skeleton is produced by scraping injuries which result from physical processes (e.g. abrasion), human activities (fin and anchor scraps), and predation (e.g. parrot fish). The loss of extensive amounts of tissue and skeleton is the removal of bumps or branches from colonies and is due to predation by excavating fish (e.g. *Bolbometopon* sp.), physical disturbances (e.g. storms and cyclones) and human activities such as diving and anchoring (e.g. Woodley et al., 1981; Glynn, 1990; Craik et al., 1990). For a given injury type, the morphological characteristics of a species (e.g. shape and tissue depth) can affect the appearance of an injury (Glynn, 1990). For example, although the outcome of feeding by some parrot fish is the superficial loss of tissue and skeleton from a colony, variations in colony shape result in different wound manifestations: gauging wounds are

produced in massive species, branchlets are lost in tabular corals, and tips are removed in arborescent corals (Glynn, 1990). The second major objective of this study is two-fold, firstly to examine how species respond to different types of injuries, and secondly, to investigate the impact of a given injury on species with different morphologies by monitoring their response (regenerative ability) to that injury.

The position of an injury within a colony is again dependent on the causative agent, and affects rates of regeneration in bryozoans (Palumbi and Jackson, 1982), the gorgonian *Plexaura homomalla* (Wahle, 1983), and some scleractinian corals such as *Acropora palmata* (Meesters and Bak, 1995), but not others (e.g. *Porites astreoides*, Meesters et al., 1993). Reduced light levels, polyp density, senescence, and substratum interactions have all been proposed to explain variations in the regeneration of lesions located at different positions within a colony (Jackson, 1979; Palumbi and Jackson, 1982; Darke, 1991; Meesters et al., 1993; Meesters and Bak, 1995). Preferential regeneration of one injury versus another within colonies of some species may have direct consequences for their population dynamics. For example, Meesters and Bak (1995) showed that the regeneration of proximal injuries was much slower than distal injuries on branches of *Acropora palmata*, and suggested that this pattern of recovery may promote asexual propagation by fragmentation. The third major objective of this study is to determine if there was any variation in the rate of regeneration of injuries located at different positions within a colony.

The response of corals to injury is important ecologically because injury reduces colony fitness in three ways. Firstly, regeneration requires energy so that resources may be diverted from growth and reproduction (e.g. Kobayashi, 1984; Rinkevich and Loya, 1989; Meesters et al., 1994; Van Veghel and Bak, 1994). Secondly, colony survival may be jeopardised since injuries provide sites for the entry of pathogens and bioeroders and space for the settlement of other organisms such as algae, sponges, and other corals (Bak et al., 1977). These organisms may later compete with the coral for food and space, or cause structural damage to the coral skeleton (Wahle, 1983; Hughes and Jackson, 1985; Babcock, 1991). Thirdly, injuries reduce the surface area available for feeding, photosynthesis and reproduction (e.g. Jackson and Palumbi, 1979; Wahle, 1983; Hughes and Jackson, 1985). A reduction in surface area may also alter colony survivorship and reproduction since both are size-specific in scleractinian corals (e.g. Hughes and Jackson, 1985; Babcock, 1991; Hall and Hughes, 1996). Understanding the processes associated with injury and regeneration in corals has also become very important from a management perspective because of the escalating degradation of coral reefs by human activities (e.g. Brown, 1988; Craik et al., 1990; Hughes, 1994). With the exception of studies which have being carried out predominantly in the Carribbean (see Table 1), there is a general lack of information on the regenerative abilities of scleractinian corals through field experimentation. This study is the first to present experimental data on the regenerative abilities of eleven species of scleractinian corals from the Great Barrier Reef

2. Methods

This study was conducted at Lizard Island (12°40′; 145°28′), a continental island on

Table 1
A list of studies showing the places and scleractinian corals for which experimental studies on injury and regeneration have been conducted

Species	Reference
Carribbean	
Acropora palmata	Bak, 1983; Meesters et al., 1993; Meesters and Bak, 1995
Montastrea annularis	Bak et al., 1977; Meesters et al., 1993, 1994 Van Veghel and Bak, 1994
Meandrina meandrites	Meesters and Bak, 1993
Porties asteroides	Bak and Steward-Van Es, 1980; Meesters et al., 1993
Agaricia agaricites	Bak et al., 1977; Bak and Steward-Van Es, 1980
Siderastrea siderea, Diploria strigosa	Meesters et al., 1993
Red Sea	
Stylophora pistillata	Loya, 1976; Rinkevich and Loya, 1989
Japan	
Acroporaformosa, A. nasuta	Kobayashi, 1984
Western Australia	
Pocillopora damicornis	Ward, 1995
Great Barrier Reef, Australia	
Pocillopora damicornis, Goniastrea retiformis, Porites mayeri, P. lichen, P. australiensis, Acropora robusta, A. millepora, A. cytherea, A. palifera A. hyacinthus, A. gemmifera	This study

the Great Barrier Reef, approximately 25 km off the eastern coast of mainland Australia. Experiments were set up at two locations, on the reef crest at North Reef (experiment 1), and in the lagoonal back-reef of Osprey Island (experiment 2). These sites were chosen because they provided a wide selection of species and an abundance of large colonies (see Table 2 for average colony sizes). One hundred and ten colonies were selected haphazardly for the experiments and individually tagged. Colonies with injuries comprising greater than 25% of the total surface area were excluded. Three of the four injuries types were experimentally inflicted on colonies to similate natural injuries. These included extensive tissue loss (tissue injury), superficial tissue and skeleton loss (scraping injuries) and extensive tissue and skeleton loss (branch removal). Tissue was removed from colonies with compressed air which was delivered through the small nozzle (5-mm diameter) of an air gun. To standardise the injury procedure for all colonies, the air pressure of the gun was maintained at 60 psi, and the nozzle was held at a distance of 3 cm away from the colony. All visible tissue was removed from the injured areas (see Bak et al., 1977). A template made of water-proof paper was used to obtain an injury of a specific area by protecting the surrounding healthy tissue from damage. This procedure left the underlying skeleton of the colony relatively intact (i.e. there was no visible damage to the architecture of the skeleton). A hammer and chisel was used to remove tissue and skeleton (scraping injury) and break off branches (branch

Table 2
Mean colony diameter±SE (cm) for the study species. Values are based on measurements taken of the maximum diameter and the diameter perpendicular to the maximum diameter for each colony, ($n = 20$)

Species	Mean diameter±SE
Goniastrea retiformis	21±2
Acopora hyacinthus	68±7
A. robusta	51±6
A. palifera	21±2
A. gemmifera	23±2
A. millepora	35±3
A. cytherea	47±4
Porites lichen	26±4
P. mayeri	21±2
P. australiensis	50±4
Pocillopora damicornis	18±1

removal). Recovery of injuries was quantified by either recording the dimensions of the lesion with a tape measure or digitising the lesion area from photographs of consecutive censuses. These data were then expressed as percentage of injury regenerated (as per Meesters et al., 1993).

2.1. Experiment 1: Interspecific differences in recovery and injury position within colonies

To determine if recovery from a tissue injury differed between species, or position within a colony, I inflicted a 4-cm^2 tissue wound at two positions (edge and centre) within 10 replicate colonies of each of seven species. Common reef crest species with contrasting morphologies were chosen and included *Acropora robusta*, *A. hyacinthus*, *A. palifera*, *Pocillopora damicornis*, *Porites mayeri*, *P. lichen* and *Goniastrea retiformis*. The growth forms of the first four species are arborescent, tabular, submassive and bushy respectively, while the last three species are massive. For the massive, tabular and bushy corals, the injuries were located at the center and edge of the colony while for the arborescent and submassive species, injuries were located at the base (centre) and tips (edge) of branches and columns. Seventy-four days after injuring the colonies, the dimensions (length and width ±1 mm) of each lesion (amount of injured area not regenerated) was measured using a tape measure. The area of the remaining lesion was then calculated and subtracted from the original injury size to estimate the amount of recovery.

2.2. Experiment 2: Interspecific differences in recovery and injury types

2.2.1. (a) Tissue removal versus scraping injuries in massive and tabular corals

To examine the response of corals to a scraping versus a tissue injury, and to investigate the impact of a scraping injury on species with different morphologies, each of 10 replicate colonies were inflicted with both injury types, the size of the injury being 4-cm^2 (Fig. 1). Two abundant coral species, the table coral *Acropora cytherea* and the

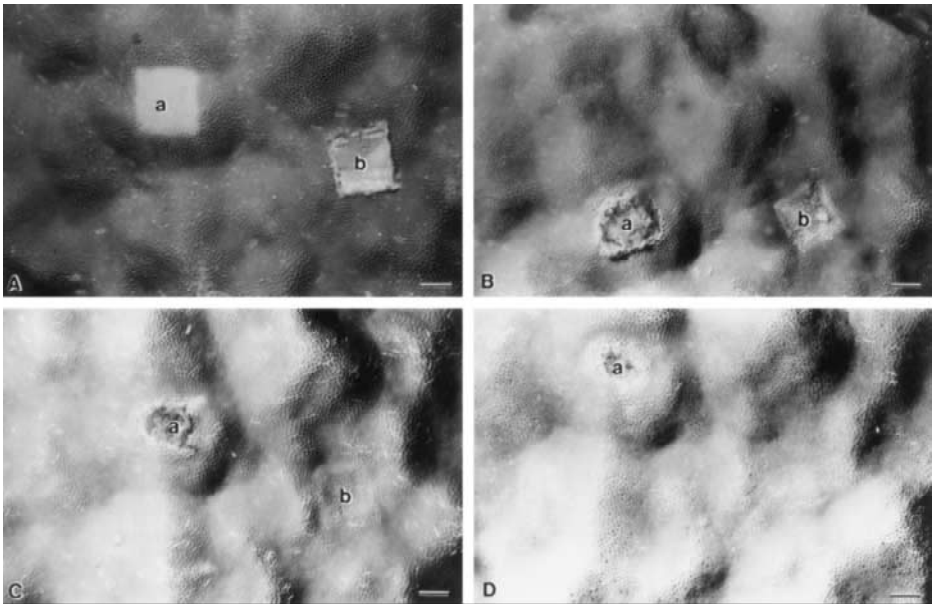

Fig. 1. A photographic series showing the recovery of the tissue and scraping injuries inflicted on *Porites* over time, (where a = tissue injury and b = scraping injury). A, Infliction of the injury in August. B, Recovery of injuries after 71 days. Note that the scraping injury has nearly completely regenerated while the unrepaired section of both injuries is overgrown with algae. C, Complete recovery of the scraping injury and partial recovery of the tissue injury. D, The tissue injury is still only partially recovered after 215 days. Scale bar = 1 cm.

massive coral *Porites australiensis* were examined to detect any interspecific differences in regeneration. Tissue was removed with compressed air as described previously, while the scraping injury was created using a hammer and chisel. For *P. australierisis*, the scraping injury was inflicted by chiselling away tissue and skeleton to a depth of 2 mm resulting in the partial injury of polyps within the injury site (average tissue depth in *Porites* is 5 mm, Barnes and Lough, 1992). For *A. cytherea*, 10–15 branchlets, 2 cm in height, were removed at their base with the chisel to create a 4-cm^2 projected area injury. The scraping injuries differed between *A. cytherea* and *P. australiensis* because they were simulating the type of injuries generated by some parrotfish. Injuries were located in the central section of the colony to avoid edge effects, and one of each type of injury was inflicted per colony. Each injury type was separated from the other by a distance of at least 5 cm. Recovery was monitored photographically over time and recorded as the amount of tissue produced around the injury.

2.2.2. (b) Tissue loss versus branch removal in branching corals

Branching acroporids on the Great Barrier Reef are prone to injuries (e.g. predation, physical disturbances) which result in either the removal of branches or tissue loss. To determine how branching corals respond to branch removal and tissue loss, and to detect interspecific differences in their response to these injury types, I removed the tissue from

one branch and broke off another in ten colonies of two species of corals. These species, *Acropora millepora* and *A. gemmifera* have different branching morphologies, corymbose and digitate respectively. Branch removal resulted in a small injury at the base of the branch, the recovery of which was monitored in addition to the regeneration of the two injury types. The dimensions of the removed branch, exposed basal area, and tissue denuded branch were measured and used to calculate branch volume lost and injury area from appropriate geometric formulas (cone shape for *A. gemmifera* branches and cylinder plus a cone top for *A. millepora* branches, see Tuma, 1979 for formulas). Both injuries were located within the centre of each colony and separated by a distance of 5 cm. Recovery was measured at 74 days. Due to the loss of 2 tags, 8 out of 10 colonies were censured for *A. gemmifera*. Regeneration was defined as the growth of a new layer of polyps over the injury site for tissue removal and the exposed area at the base of the branch, while recovery of the branch was defined as the volume of new growth. The amount of new vertical growth for both tissue injury and branch removal was also recorded to allow comparisons between injury types.

2.3. Statistical analysis

An analysis of variance was used to test for differences between species and injury positions or types. A Least Significant Difference Test (LSD) was used to determine differences between significant factors within the anova model. Data not conforming to the assumptions of anova were tested using a Kruskal-Wallis one-way nonparametric analysis of variance.

3. Results

3.1. Experiment 1: Interspecific differences in recovery and injury position within colonies

Striking differences were detected in the regenerative ability of corals (Fig. 2, Table 3). In general, the branching species regenerated a greater proportion of their injuries than the massive and submassive species within 74 days (Fig. 2). Specifically, *A. robusta* regenerated approximately 1.5 times more of the injured area within 74 days then *P. damicornis* and *A. hyacinthus*, which in turn, regenerated 4–5 times more of their injuries than did *P. lichen*, *G. retiformis*, *P. mayeri* and *A. palifera* (LSD test). The position of the injury did not affect the rate of regeneration in *A. robusta*, *A. hyacinthus*, *P. damicornis*, *P. lichen*, *G. retiformis* and *A. palifera* (LSD test). In contrast, *P. mayeri* regenerated 25% of its central injuries but none of the edge injuries within 74 days (Fig. 2).

The removal of tissue with compressed air resulted in the death of tissue within the template area for all species, as evidenced by the subsequent colonisation of the injury site by algae. Consequently, recovery of the injury involved the overgrowth of the algae with new polyps. Two different modes of regeneration were observed for the tissue injury. Regeneration in *Acropora* species was characterised by the production of a

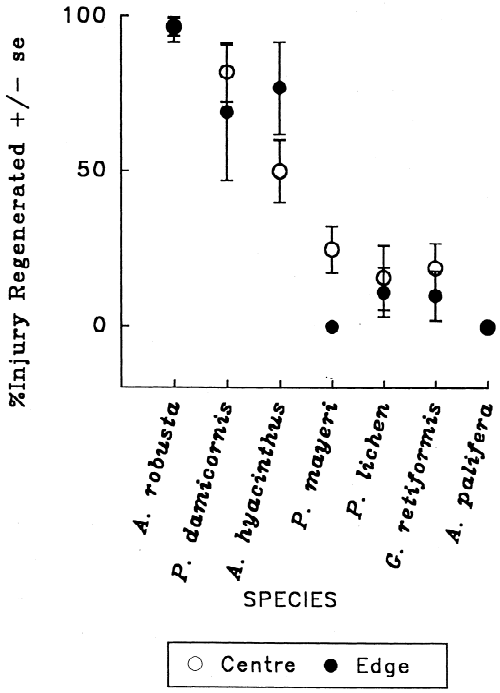

Fig. 2. The amount of regeneration of a central and edge injury for several common reef crest species over 71 days. The error bars are one standard error.

smooth band of tissue and skeleton which enveloped the injury and grew over the colonising algae. Polyps were than produced along the margin of the band that was closest to the uninjured tissue. In contrast, rather then producing a smooth band, the massive species (*Porites lichen*, *P. mayeri* and *G. retiformis*) and *Pocillopora damicornis* produced new polyps directly around the margin of the injury.

3.2. Experiment 2: Interspecific differences in recovery and injury types

Marked differences were detected in the recovery of injuries inflicted on *Acropora cytherea* and *Porites australiensis*. Repair was greater for tabular *A. cytherea* than for

Table 3
Analysis of variance table for the comparison of the amount of regeneration between species and injury position for experiment 1 at North Reef, Lizard Island

Source	*df*	*F* ratio	*P*
Species	6	54.48	0.0000
Injury Position	1	0.16	0.6858
Species*Injury Position	6	2.19	O.0482
Error	122		

the massive, *P. australiensis* (Fig. 3). *Acropora cytherea* showed almost complete recovery from both injury types (i.e. scraping injury and tissue loss) by the first census at 71 days (Fig. 3) while *P. australiensis* repaired the scraping injury much faster than the tissue injury, with recovery being complete within 153 and 286 days respectively (Figs. 1 and 3). For both injury types, *P. australiensis* showed a decrease in regeneration rate with time (Fig. 3). The mode of regeneration of these species was similar to that described previously (experiment 1), with *A. cytherea* responding the same as the other *Acropora* species and *P. australiensis* responding similarly to the other massive species.

Striking differences were also detected in recovery of tissue loss versus branch removal for *A. millepora* and *A. gemmifera* (Table 4). The amount of vertical extension of tissue along the branch was approximately 10 times greater than the vertical extension

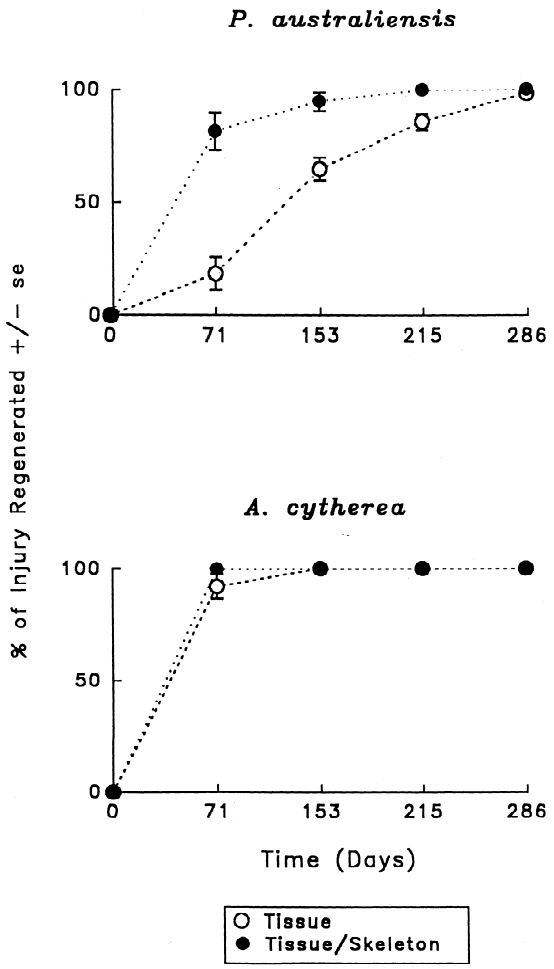

Fig. 3. Percentage of injury regeneration over time for *P. australiensis* and *A. cytherea*. The error bars are one standard error.

Table 4
(a) Linear extension of new growth for the two injury types (tissue removal and branch removal) inflicted on *A. millepora* and *A. gemmifera* in experiment 2b (mean±one standard error). (b) Injury size and regeneration (percent of injury recovered) for the three injury types inflicted on *A. millepora* and *A. gemmifera* in experirnent 2b (mean±one standard error). (Number of colonies = 8 and 10 for *A. gemmifera* and *A. millepora* respectively)

(a)				
Injury type	*A. gemmifera*		*A. millepora*	
	Regeneration	Statistic	Regeneration	Statistic
Linear extension of tissue (mm)	3.19±0.86	KW = 14.4	4.3±0.31	KW = 11.3
Linear extension of branch (mm)	0.25±0.13	$P = 0.0001$	0.55±0.12	$P = 0.0008$
(b)				
Injury type				
Tissue removal (cm^2)	49±23	25±2	68±19	21±2
Branch removal (cm^3)	3.5±1.2	7±0.7	7.0±2.2	6±0.9
Exposed area at branch base (cm^2)	100	2.92±0.2	100	0.98±0.1

KW represents the Kruskal-Wallis statistic obtained by comparing the differences in linear extension between the two injury types, and P equals the probability.

of a new branch (Table 4a). This result appears to be related to differences in the patterns of regeneration observed between injury types. Regeneration of the tissue injury in the *Acopora* species involved the sealing of the area with a fine, porous layer of tissue and skeleton, followed by the development of polyps within this area. Branch replacement also involved a number of intergrading stages. Firstly, the base of the branch where the breakage had occurred was sealed by a thin layer of skeleton and tissue. Secondly, a central axial polyp and several radial polyps then developed within this thin layer and thirdly, vertical extension of the branch occurred with the growth of the axial polyp. The initial extension of the axial polyp produced a narrow branch (0.3-cm diameter) which gradually thickened at the base.

The pattern of regeneration of a given injury type was similar for *A. millepora* and *A. gemmifera* (Table 4b), although some inter-colony variations were observed. All colonies for both species showed some degree of regeneration of the tissue injury (percent of injury recovered was 68±19% and 49±23% for *A. millepora* and *A. gemmifera* respectively) after 74 days (Table 4b). Additionally, they both rapidly repaired the injured area at the base of the branch where breakage occurred, completely sealing the area in less than 74 days (Table 4b). However, within this sealed area, there was some variation in polyp development between colonies for *A. gemmifera* with 25% of colonies displaying no signs of polyp development by the end of the experiment. Conversely, all *A. millepora* colonies showed some polyp development within the 74 day period. Some variations in the number of colonies of each species exhibiting vertical branch extension was also detected. In *A. gemmifera*, 3 out of 8 colonies showed some signs of vertical extension, and replaced an average of 7±0.7% of the original branch

volume. On the other hand, the majority of *A. millepora* colonies (8 out of 10) extended vertically, showing an average of 10.4±6.7% replacement of the original branch volume. Although the overall pattern of regeneration of the different injury types was similar for *A. gemmifera* and *A. millepora*, regeneration was faster in *A. millepora* than *A. gemmifera* (Table 4b). This was probably due to the fact that the injuries inflicted on *A. gemmifera* were larger (Table 4b).

4. Discussion

Regenerative ability in this study differed both between species for a specific injury and between different injury types for a particular species. Generally, colonies of branching species regenerated more of their injury within a given time period than did the massive species (Figs. 2 and 3 Table 4). Furthermore, rankings of the recovery of the different types of injuries are as follows: scraping injury > tissue injury > branch removal (Figs. 2 and 3 Table 4). The impact that an injury has on a particular species seems to be reflected in the ability of the colony to regenerate.

4.1. Interspecific differences in regeneration

Jackson (1979) proposed that colonial organisms with morphologies that showed an increasing commitment to their place of settlement would invest more resources into defence and maintenance as opposed to those morphologies which were more fugitive in nature. A ranking of the regenerative ability of the species examined in this study show a similar result to that proposed by Jackson (1979): arborescent (tree) ≥ bushy (tree) ≥ tabular (plate) > massives (mounds) > submassive (sheet). The impact that a particular injury has on a coral is related to tissue and skeleton volume and tissue area. The amount of tissue covering a colony infers something about its potential for feeding, respiration, and reproduction as well as offering a means of defence against biotic and abiotic processes through allelochemistry, nemocyst discharge, mucus production, and so on. In corals, some species are fleshier than others (Veron, 1986), so that the impact of a tissue injury will depend on the depth of tissue covering the colony and implies that the deeper the tissue the greater the loss. In this study, the regeneration of these tissue injuries was slower for massive species than for branching species and may be related, in part, to tissue depth which is generally greater in massive corals. However, since the repair of tissue injuries usually involves the production of both skeletal and tissue matter, the amount of investment in skeletal material will also be important. Although there is a great deal of interspecific variation in the density of coral skeletons, skeletal densities can usually be associated with different growth forms (Hughes, 1987). For example, many branching *Acroporas* have a pronounced axial gradient in skeletal density, with the base being much denser than the tips (Hughes, 1987). In these species, the mode of regeneration of small injuries is distinctive. The injured area is rapidly sealed with a fine, porous layer of tissue and skeleton, followed by the development of polyps within this area (Bak, 1983, this study). This mode of recovery suggests that the skeleton produced in regenerating areas has a similar density to that of the faster growing tips. In contrast

to the branching *Acroporas*, the submassive *A. palifera* does not have a pronounced axial gradient in skeletal density but maintains a fairly dense column from tip to base (average density $= 1.9 \pm 0.12$, $n = 6$). Additionally, it did not regenerate any of its injured area within 74 days. This infers that the production of such a dense skeleton during regeneration may incur a large drain on resources and slow recovery in *A. palifera* colonies. Similarly, differences in the skeletal investments of *A. millepora* and *A. gemmifera* influence their regeneration rates. The volume of tissue and skeleton lost in branch removal was 16% less for *A. millepora* than for *A. gemmifera*, while the amount of regeneration after 74 days was approximately double (Table 4).

4.2. Injury characteristics

The characteristics of an injury play a major role in establishing rates of regeneration because they determine the amount of damage which is inflicted on a colony. Marked differences were detected in the regeneration rates of the three different injury types, tissue, scraping, and branch removal. For example, the recovery of a tissue injury versus branch removal in colonies of *A. millepora* and *A. gemmifera* varied within a period of 74 days (Table 4), with vertical extension of new tissue being approximately 10 times faster than branch extension. This infers that the loss of tissue from a branch has less impact on a colony than the removal of a branch. This may be due to the fact that the amount of material required to replace the lost tissue is much less than that needed to replace a branch since there is less investment required for skeletal material (i.e. the framework of the branch is left intact and only a thin layer of porous skeleton is needed for recovery). Colony survival, on the other hand, may be greater following the removal of a branch as compared with a tissue injury since the exposed branch base was rapidly sealed ($>$ 74 days, Table 4). Consequently, the potential for invasion by bioeroders or settlement by superior competitors would be greater for tissue injuries since the injured area is exposed and vulnerable for a longer period of time.

Poriles australiensis regenerated its scraping injuries more quickly than its tissue injury (Fig. 3). Similar results were obtained in the Caribbean by Bak and Steward-Van Es (1980) for *Porites asteroides*. Bak and Steward-Van Es (1980) suggested that the repair of the tissue and skeleton injury was accelerated by damage to the skeleton, and that the rapid recovery observed was assisted by regeneration of the partially injured polyps within the injury site. I propose that the difference in regeneration rates between scraping and tissue injuries may also be related to mucus production. In scraping injuries, mucus production by the partially injured polyps may inhibit algal settlement onto the injury site, allowing regeneration to proceed without interference from competition. In contrast, it is not possible for mucus to be produced from sites where the tissue has been removed since the polyps have been killed. Consequently, the site of a tissue injury is colonised by algae and some of the resources available for regeneration will be diverted towards competition, resulting in slower regeneration rates.

Contrary to several previous studies, (e.g. Palumbi and Jackson, 1982; Wahle, 1983; Meesters and Bak, 1995), the location of an injury within a colony did not affect regeneration for six out of the seven species investigated (LSD, Fig. 2). Only *Porites mayeri* showed a significant difference, with the central injury regenerating 25% of its

area within 14 days while no recovery was recorded for the edge injury over the same time (Fig. 2). Variations in the regeneration of lesions located at different positions within a colony have been related to reduced light levels, polyp density, senescence, and substratum interactions (Jackson, 1979; Palumbi and Jackson, 1982; Darke, 1991; Meesters et al., 1993; Meesters and Bak, 1995). Meesters et al. (1993) demonstrated that the regeneration of artificial lesions on *Acropora palmata* and *Montastrea annularis* colonies was slower in high sediment areas and suggested that this pattern was partially due to a decline in irradiance. *Diploria strigosa*, *Siderastrea siderea* and *Porites astreoides*, on the other hand, did not show a difference in the recovery of lesions in high and low sediment sites nor was there a difference in the rate of repair of lesions located on the top and side of *P. astreoides* colonies, even though light was reduced by 70% at the sides. Many corals survive in low light levels by photoadapting (Falkowski et al., 1990). This phenomena may explain why six out of the seven species in this study did not show a difference in the regeneration of lesions at the edge and center of their colonies. In contrast, *Porites mayeri* may have a low capacity to photoadapt. Alternately, the difference in regeneration between these two positions in *P. mayeri* may be related to differences in polyp density. Darke (1991) demonstrated that there was a 10% decline in the density of polyps from the summit to the base of *Porites* colonies, and suggested that higher polyp density reflected relatively higher rates of tissue growth to skeletal growth. Consequently, in *P. mayeri*, lower polyp densities at the base of the colony may hinder regeneration in that area. In contrast to *P. mayeri*, regeneration of edge and central injuries was not significantly different for. *Porites lichen* (Fig. 2 Table 1). These two species may differ in this respect because *P. mayeri* buds extra-tentactularly while *P. lichen* buds intra-tentactularly (Veron and Pichon, 1982). Studies on patterns of polyp density in massive corals that bud intra-tentactularly have yet to be done. Senescence and substratum interactions have also been proposed to explain variations in the regeneration of injuries at different locations within the colony (Jackson, 1979; Meesters and Bak, 1995). However, these processes are unlikely explanations here because substratum interactions would not only affect *P. mayeri* but the other massive species as well, and senescence is most likely to occur in colonial species such as *Steginoporella* sp. and *Acropora palmata* that show a proximal to distal gradient in regeneration and growth (Palumbi and Jackson, 1982; Meesters and Bak, 1995).

5. Conclusions

Clearly, the impact that an injury has on a colony is determined by its regenerative ability, which in turn, is greatly influenced by the morphological characteristics of a species. Jackson (1979) predicted that the amount of resources invested in maintenance and defence would vary in sessile animals with different morphologies, and predicted that species with a high commitment to their place of settlement would invest more in maintenance and defence. The results of this study support this hypothesis since the ranking of regenerative ability of the different morphologies was arborescent > bushy > tabular > massive > submassive. Poor regenerative ability in corals may lead to reduced colony fitness since regeneration potentially diverts resources away from growth and

reproduction, reduces survival, reproductive output, feeding and photosynthetic capacity, and destroys stored reserves (e.g. Bak et al., 1977; Wahle, 1983; Rinkevich and Loya, 1989).

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References

Babcock, R.C., 1991. Comparative demography of three species of scleractinian corals using age- and size-dependent classifications. *Ecol. Monog.*, Vol. 1, pp. 225–244.

Bak, R.P.M., 1983. Neoplasia, regeneration and growth in the reef building coral *Acropora palamata*. *Mar. Biol.*, Vol. 77, pp. 221–227.

Bak, R.P.M. and Y. Steward-Van Es, 1980. Regeneration of superficial damage in the scleractinian corals *Agaricia agaricites F. purpurea* and *Porites astreoides*. *Bull. Mar. Sci.*, Vol. 30, pp. 883–887.

Bak, R.P.M., J.J.W.M. Brouns and F.M.L. Heys, 1977. Regeneration and aspects of spatial competition in the scleractinian corals *Agaricia agaricites* and *Montastrea annularis*. *Proc. 3rd. Int. Coral Reef Symp., Miami*, pp. 143–148.

Barnes, D. and J. Lough, 1992. Systematic variations in the depth of skeleton occupied by coral tissue in massive colonies of *Porites* from the Great Barrier Reef. *J. Exp. Mar. Biol.*, Vol. 159, pp. 113–128.

Brown, B.E. 1994. Solar damage in intertidal corals. *Mar. Ecol. Prog. Ser.*, Vol. 105, pp. 219–230.

Brown, B.E., 1988. Assessing environmental impacts on coral reefs. *Proc. 6th Int. Coral Reef Symp. Townsville*, Vol. 1, pp. 71–80.

Brown, B.E. and L.S. Howard, 1985. Assessing the effects of 'stress' on reef corals. *Advances in marine biology, Vol. 22.*, Academic Press Inc., London, pp. 1–63.

Chadwick-Furman, N.E., 1995. Effects of scuba diving on coral reef invertebrates: relative vulnerabilities and critical levels of diving activity. *Proc. 6th Int. Conf. Coelenterate Biology, Amsterdam* (in review).

Connell, J.H., 1973. Population ecology of reef building corals. In, *Biology and geology of coral reefs*, 2, edited by O.A. Jones and R. Endean, Academic Press, New York, pp. 205–245..

Craik, W., R. Kenchington and G. Kelleher, 1990. Coral-Reef Management. In, *Ecosystems of the world, coral reefs*, edited by Z. Dubinsky, Elsevier Science Publishers, N.Y. pp. 453-467.

Darke, W.M., 1991. Growth and growth form of the massive coral *Porites*. PhD thesis, James Cook University of North Queensland, Australia.

Falkowski, P.G., P.L. Jokiel and R.A. Kinzie, 1990. Irradiance and corals. In, *Ecosystems of the world coral reefs*, edited by Z. Dubinsky, Elsevier Science Publishers, N.Y. pp. 89–107.

Glynn, P.W., 1990. Feeding ecology of selected coral-reef macroconsumers: Patterns and effects on coral community structure. In, *Ecosystems of the world, coral reefs*, edited by Z. Dubinsky, Elsevier Science Publishers, N.Y. pp. 439–452.

Grigg, R.W. and S.J. Dollar, 1990. Natural and anthropogenic disturbance: on coral reefs. In, *Ecosystems of the world, coral reefs*, edited by Z. Dubinsky, Elsevier Science Publishers, N.Y. pp. 365–400.

Hall, V.R. and T.P. Hughes, 1996. Reproductive strategies of modular organisms: comparative studies of reef building corals. *Ecology*, Vol. 77, pp. 950–963.

Hughes, T.P., 1987. Skeletal density and growth form of corals. *Mar. Ecol. Progr. Ser.*, Vol. 35, pp. 259–266.

Hughes, T.P., 1989. Community structure and diversity of coral reefs: The role of history. *Ecology*, Vol. 70 (1), pp. 275–279.

Hughes, T.P., 1994. Catastrophes, phase shifts, and large-scale degradation of a Caribbean reef. *Science*, Vol. 265, pp. 1547–1551.

Hughes, T.P. and J.B.C. Jackson, 1985. Population dynamics and life histories of foliaceous corals. *Ecol. Monog.*, Vol. 55, pp. 141–166.

Jackson, J.B.C., 1979. Morphological strategies of sessile animals. In, *Biology and Systematics of Colonial Organisms*, edited by G. Larwood and B.R. Roser, Academic Press, London. pp. 499–555.

Jackson, J.B.C. and S.R. Palumbi, 1979. Regeneration and partial predation in cryptic coral reef environments: Preliminary experiments on sponges and ectoprocts. *Colloq. Int. C.N.R.S.*, Vol. 291, pp. 303–308.

Kobayashi, A., 1984. Regeneration and regrowth of fragmented colonies of the hermatypic corals *Acroporaformosa* and *Acropora nasuta*. *Galaxea*, Vol. 291 pp. 13–23.

Loya, Y., 1976. Skeletal regeneration in a Red Sea Scleractinian coral population. *Nature*, Vol. 261, pp. 490–491.

Meesters, E.H. and R.P.M. Bak, 1993. Effects of coral bleaching on tissue regeneration potential and colony survival. *Mar. Ecol. Prog. Ser.*, Vol. 96, pp. 189–198.

Meesters, E.H. and R.P.M. Bak, 1995. Age-related deterioration of a physiological function in the branching coral *Acropora palmata*. *Mar. Ecol. Proc. Ser.*, Vol. 121, pp. 203–209.

Meesters, E.H., A. Bos and G.J. Gast, 1993. Effects of sedimentation and lesion position on coral tissue regeneration. *Proc. 7th Int. Coral Reef Symp., Guam,* pp. 681–688.

Meesters, E.H., M. Noordeloos and R.P.M. Bak, 1994. Damage and regeneration: links to growth in the reef building coral *Montastrea annularis*. *Mar. Ecol. Proc. Ser.*, Vol, 112, pp. 119–128.

Palumbi, S.R. and J.B.C. Jackson, 1982. Ecology of cryptic coral reef communities. II. Recovery from small disturbance events by encrusting bryozoa: The influence of 'host' species and lesion size. *J. Exp. Mar. Biol. Ecol.*, Vol. 64., pp. 103–115.

Rinkevich, B. and Y. Loya, 1989. Reproduction in regeneration colonies of the coral *Stylophora pistillata*. In, *Enviromental quality and ecosystems stability: Vol. IV-B, Environmental Quality*, edited by E. Spanier, Y. Steinberger, N. Luria, ISEEQS, Israel.

Stafford-Smith, M.G., 1993. Sediment-rejection efficiency of 22 species of Australian scleractinian corals. *Mar. Biol.*, Vol. 115, pp. 229–243.

Tuma, J.J., 1979. Engineering Mathematics Handbook. Definitions, theorems, formulas and tables. McGraw-Hill Book Company, N.Y.

Van Veghel, M.L.J. and R.P.M. Bak, 1994. Reproductive characteristics of the polymorphic Carribbean reef building coral *Montastrea annularis*. III. Reproduction in damaged and regenerating colonies. *Mar. Ecol. Prog. Ser.*, Vol. 109, pp. 229–233.

Veron, J.E.N., 1986, *Corals of Australia and the Indo-Pacific*, University of Hawaii Press, USA.

Veron, J.E.N. and M. Pichon, 1982. Scleractinia of Eastern Australia. Part IV: Family Poritidae. *Australian Institute of Marine Science Monographs Series*, Vol. 5., Australian Institute of Marine Science, Australia.

Wahle, C.M., 1983. Regeneration of injuries among Jamaican gorgonians: The roles of colony physiology and environment. *Biol. Bull.*, Vol. 164, pp. 778–790.

Ward, S., 1995. The effect of damage on the growth, reproduction and storage of lipids in the scleractinian coral *Pocillopora damicornis* (Linnaeus). *J. Exp. Mar. Biol. Ecol.*, Vol. 187, pp. 193–206.

Woodley, J.D., E.A. Chornesky, P.A. Clifford, J.B.C. Jackson, L.S. Kaufinan, N. Knowlton, J.C. Lang, M.P. Pearson, J.W. Porter, M.C. Rooney, K.W. Rylaarsdam, V.J. Tunnicliffe, C.M. Wahle, J.L. Wulff, A.S.G. Curtis, M.D. Dallmeyer, B.P. Jupp, M.A.R. Kodhl, J. Neigel and E.M. Sides, 1981. Hurricane Allen's Impact on Jamaican Coral Reefs. *Science*, Vol. 214, pp. 749–755.